

What Should the Agent Withhold? Designing Cognitive Friction Against Over-Reliance in Visualization Design Learning

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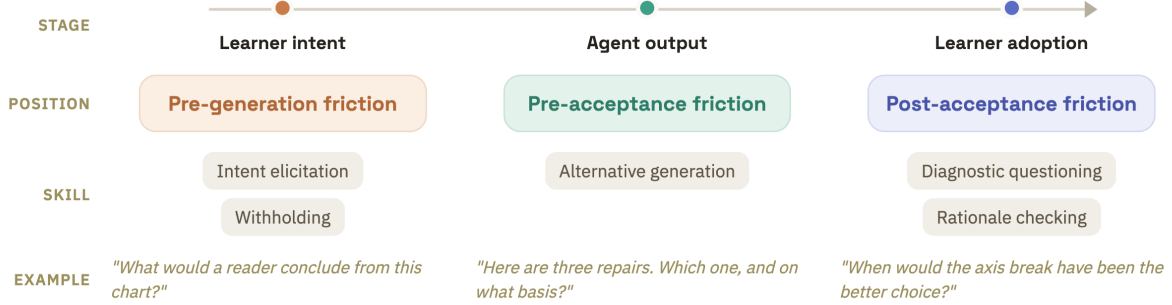


Figure 1: Cognitive friction intervenes at three points in the learner-agent loop during visualization design: before the agent generates output, before the learner accepts it, and after adoption. Each position requires a different agent capability to deliver, shown below the corresponding position. The example prompts are adapted from the case study in Section 5. Rationale checking can also gate pre-acceptance (see Table 2).

ABSTRACT

Students learning visualization design increasingly use GenAI-powered authoring tools, which can produce a chart from a short natural-language description. While these tools lower the barrier to producing visualizations, they also allow learners to produce a satisfactory chart without making the design decisions the exercise was intended to develop, which can give an illusion of competence while leaving design skills underdeveloped. In this position paper, we make a case for AI agents that facilitate cognitive friction as a strategy to support visualization design learning. We define friction for this setting, distinguish it from the incidental cost of a difficult interface, and organize it by the point in the learner-agent loop at which it intervenes, yielding three positions. We then describe five skills an agent needs to deliver friction at these positions. We illustrate both the positions and the skills through a chart-critique exercise suited to introductory visualization teaching. We then discuss what would be needed to test whether an agent with these skills actually supports learners’ design reasoning.

Index Terms: Visualization education, generative AI, cognitive friction, design judgment, over-reliance.

1 INTRODUCTION

Generative AI has changed how visualizations are produced. Tools such as Data Formulator 2 [24] accept a description of what a learner wants to compare, infer the data transformation that description implies, and render the result, while systems such as LIDA [7] generate candidate visualizations from a dataset with no specification supplied. For practitioners these tools remove work that was less creative and rather routine, such as data processing or creating an initial sketch. For students in a visualization course, the same re-

moval is rather disruptive, because the work being removed is often the work the course exists to teach.

Visualization design is not a matter of applying a rule to reach a single correct chart. Parsons and colleagues characterize it as a process in which problem framing and solution develop together, so that what a visualization is for becomes clearer through repeated attempts to make it [20], and they identify design judgment as the capability that distinguishes experienced practitioners [18]. Judgment of this kind is developed through decisions: deciding which comparison the chart should support, choosing an encoding that serves it, and revising when the two do not align. These are precisely the decisions a capable agent can now make on a learner’s behalf. A student who submits a well-formed chart may therefore have exercised the relevant reasoning, or may have delegated all of it, and the artifact does not distinguish the two cases. This is why an engaged, reflective process matters for evaluating learning, not only a correct result, and why the traces such a process leaves behind become part of what should be assessed.

There is reason to think the second case is common. Bućinca et al. found that users of AI recommendation systems accept output without deliberation unless the interface actively interrupts them, and that they prefer the interfaces on which they perform worst [5]. Vasconcelos et al. explain this behavior as a cost-benefit judgment, in which engaging with an output is skipped when its expected return does not justify the effort [23]. In education specifically, Bastani et al. found in a randomized trial with approximately one thousand secondary students that access to an unrestricted GenAI tutor improved performance while the tool was available but reduced independent examination scores by roughly seventeen percent, and that the harm disappeared when the same model was configured to withhold direct answers [3]. Kazemitabaar et al. report a comparable pattern in programming education. They show that interfaces requiring engagement with AI-generated code can mitigate it [11].

These findings suggest a response. We can deliberately reintroduce some of the effort the tool removes. Prior work calls this design friction, arguing that introducing difficulty on purpose disrupts automatic acceptance and encourages more careful engagement [6].

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We adopt this idea for visualization design learning, where the effort is not incidental to the goal but part of how the goal is reached.

2 RELATED WORK

2.1 Over-reliance and friction in AI-assisted work

Concern about uncritical acceptance of AI output predates generative models. Lee and See established appropriate reliance as the target for automation design, noting that calibration can fail in either direction [14]. In the context of AI-assisted decisions, Bućinca et al. showed that cognitive forcing functions, which interrupt the flow of interaction and require the user to engage before an AI recommendation is revealed or accepted, reduce over-reliance relative to explanation-based interventions [5]. Their study also produced a result that constrains how such designs can be evaluated, in that participants rated the most effective interfaces least favourably. Vasconcelos et al. provide a mechanism for the underlying behaviour, showing that users default to acceptance when the cognitive cost of verification exceeds its perceived benefit [23]. More recently, Tankelevitch et al. have argued that generative systems place substantial metacognitive demands on users, who must formulate intentions and evaluate outputs in domains where they may lack the expertise to do either [22], and Lee et al. report from a survey of knowledge workers that higher confidence in AI output is associated with reduced critical engagement [13].

2.2 Difficulty as a design principle in learning

The claim that learning can benefit from conditions that make performance harder is well established in educational research. Bjork and Bjork describe desirable difficulties, manipulations that slow acquisition while improving long-term retention and transfer [4]. Kapur’s work on productive failure shows that learners who attempt a problem before instruction outperform those given instruction first [9], and his later analysis is equally important for design purposes, since it distinguishes productive from unproductive failure and thereby establishes that the benefit has an upper bound [10]. Most directly relevant to our argument, Reiser identifies two mechanisms through which scaffolding operates: structuring, which reduces the complexity a learner confronts, and problematizing, which deliberately draws attention to aspects of the work a learner would otherwise pass over [21]. Problematizing is the educational counterpart of what HCI calls friction, and its longstanding place in the scaffolding literature indicates that adding difficulty is a recognized instructional mechanism rather than a departure from support.

2.3 Design judgment in visualization and its assessment

Visualization education has recently been identified as an area with substantial open challenges, spanning curriculum, pedagogy, and assessment [2]. Within visualization, Parsons and colleagues have studied how practitioners actually work. They document fixation, in which designers settle prematurely on a familiar solution, and identify authoring tools that fail to support exploration of alternatives as one contributing cause [19]. Later work describes design as the co-evolution of framing and solution [20] and treats design judgment as a core professional capability [18]. Assessment instruments in the field have taken a different focus. VLAT and its variants measure the accuracy with which readers extract information from charts [15, 17], and CALVI extends this to the detection of misleading designs [8]. Both address the interpretation of existing visualizations. What a learner can produce, and on what grounds they chose to produce it, falls outside their scope. Two recent systems address related but distinct problems. VizCrit offers computational feedback via annotated overlays, focusing on how actionable feedback should be rather than when it is delivered [16]. DesignMentor guides LLM chatbots through a fixed six-stage coaching se-

quence drawn from the Cognitive Apprenticeship Model [1]. Both concern the content of feedback once a learner has engaged the system. We instead ask when friction should be introduced at all.

The capability at stake in visualization education is design judgment. Unlike an isomorphic programming task, there is no post-test that can adjudicate whether a learner has exercised it, and the design space of frictions appropriate to it has not been described.

3 COGNITIVE FRICTION IN VISUALIZATION DESIGN LEARNING

3.1 Definition

We use the term *friction*, introduced in Section 1, to mean an element of an interaction that requires the user to perform a deliberate cognitive act they could otherwise have skipped. The act is required before the agent produces an output, before the user accepts one, or, after adoption, to test the reasoning behind it. A tool that will not generate a chart until the learner has named the fields to be compared has introduced friction in this sense. A tool that is simply slow, or whose controls are hard to find, is different. It costs the learner time or patience, not thought. The learner can pay that cost without ever considering the design. The relevant precedent is Bućinca et al.’s cognitive forcing functions, interventions whose purpose is to make a user deliberate at the moment they would otherwise have accepted [5].

3.2 Three constraints on introducing friction

Before describing where friction can be introduced, we note three limits on how it should be used in visualization design learning. They are drawn from the literature reviewed above and they recur in the taxonomy that follows.

Friction varies continuously, and deployed tools already differ in how much they impose. Data Formulator 2 requires the learner to name the fields and concepts to be visualized, and it exposes the data transformation it generates as an artifact that can be inspected and edited [24]. A conversational tool that returns a finished chart from a single sentence does neither. Both are low-friction compared to writing a specification by hand, and they are not equivalent to each other. The question for design is therefore how much friction to impose and where, rather than whether to impose any.

Friction is a form of scaffolding: it helps by problematizing rather than by structuring. Reiser describes two ways an instructor can support someone: by removing complexity, so the learner can focus on the essential steps, or by deliberately drawing attention to a part of the task the learner would otherwise skip [21]. The second is what Reiser calls problematizing. Friction, as we use the term, is problematizing. An agent that adds friction has not stopped helping. It is helping in the second way instead of the first.

Friction has an upper bound, and learner preference cannot locate it. Kapur’s distinction between productive and unproductive failure [10] and Vasconcelos et al.’s cost–benefit account [23] both imply a threshold beyond which additional demand produces disengagement. This mirrors what Koedinger and Aleven call the assistance dilemma in intelligent tutoring systems. Give too much help and the learner does not engage; give too little and they cannot proceed [12]. Kazemitabaar et al. observed this directly, finding that techniques requiring repetitive effort increased frustration without improving performance, while the most effective technique engaged learners in the problem-solving itself [11]. Bućinca et al.’s finding that participants preferred the interfaces on which they performed worst [5] rules out satisfaction data as a means of calibration.

3.3 Three positions for friction

We think of a learner’s work with an agent as a sequence of exchanges: the learner states something, the agent responds, and the

Table 1: Three positions at which an agent can introduce friction while a learner designs a visualization with its assistance. Example prompts are adapted from the exercise in Section 5.

Position	What the agent withholds	Example prompt	The design decision it returns to the learner
Pre-generation	Any output, until the learner states an intent	<i>“What would a reader conclude from this chart?”</i>	What the chart is for, and which fields bear on it
Pre-acceptance	Adoption, until the learner chooses among alternatives and gives a reason	<i>“Here are three repairs. Which one, and on what basis?”</i>	Which visual channel should carry the quantity
Post-acceptance	Nothing further. The reason already given is tested	<i>“When would the axis break have been the better choice?”</i>	The conditions under which the chosen rule holds

learner acts on that response, either by generating something new, or by asking something further. We define a **position** as a moment within this exchange at which the agent can choose to introduce friction rather than proceed automatically. Figure 1 gives an overview of where friction can enter this loop and what an agent must be able to do at each point.

Several interventions have been proposed for reducing over-reliance in AI-assisted work: delaying the presentation of a suggestion [5], requiring the user to justify their acceptance of it [23], asking the user to critique an output before revealing the system’s assessment [11], and surfacing alternatives the user has not considered [19]. These have accumulated without a shared organising principle. We propose a single organizing dimension: the point in the learner–agent loop where the friction intervenes. This yields three positions, summarized in Table 1.

Pre-generation friction requires the learner to state something before the agent produces any output. What is required is typically the analytic question the chart is meant to answer, or a specification of what is to be compared. In visualization this is where framing is decided. For example, a learner who states that the chart should show whether growth is accelerating has fixed a task against which every later encoding choice can be assessed, whereas a learner who generates first receives a chart shaped by whatever question the agent inferred, and has no independent standard for judging it. Delayed suggestion belongs here, as does any requirement to sketch or specify before generating.

Pre-acceptance friction allows the agent to produce output but requires the learner to do something before adopting it. The form this takes in visualization is the presentation of genuine alternatives, meaning designs that differ in which visual channel carries the quantitative variable or in the level at which the data is aggregated, rather than in color or typeface. A learner shown a bar chart, a line chart, and a dot plot of the same measure must compare encodings in a way that a single returned chart does not require. Justification prompts and critique-first workflows also occupy this position, since both interpose a stated reason between output and adoption.

Post-acceptance friction applies after a design has been adopted, and its purpose is to establish whether the reasoning behind the choice generalizes. A learner may reject a truncated axis for the correct reason without knowing the conditions under which a broken axis is acceptable, and the submitted chart is identical in both cases. A question that asks when the rejected alternative would have been the better choice makes the difference visible.

4 SKILLS FOR A FRICTION-DELIVERING AGENT

Friction cannot be delivered by a fixed interface alone. A text box that must be filled before submission can enforce a pause, but it cannot judge whether three alternatives are meaningfully different, or whether a learner is stuck versus just avoiding effort. That judgment depends on the data, the task, and where the learner is in the exercise.

The interventions in Section 3.3 each describe what an interface should do, not what the agent behind it must judge in order to do it well. We name that judgment for five skills, offered as a working vocabulary to be refined against instructor practice (Section 6).

Intent elicitation. Before generating anything, the agent asks what the chart should say to its reader or, for an existing chart, what a reader would take from it. The judgment lies in telling a real claim from a vague topic. “I want to show sales over time” is a topic. “I want to show that growth stalled after Q2” is a claim, and only the latter gives the agent something to check every later choice against. Delayed suggestion withholds output until some condition is met, but does not say what the agent must judge in that condition [5]. Section 5 shows this in the diagnosis step.

Withholding. The agent declines to produce output that is within its capability. This is the least mechanical of the five, because its correct application is intermittent: a learner blocked on the syntax of a specification should be unblocked, whereas a learner about to accept a default encoding without considering alternatives should not be. Kazemitabaar et al.’s most effective technique withheld generated code and released it in stages as the learner articulated what each stage required [11]. This suggests withholding is most effective when paired with a route forward rather than applied as a refusal.

Alternative generation. The agent produces several designs that differ in a respect the learner must reason about. Three charts differing in palette are not alternatives in this sense. Three that place the same quantity on length, position, and angle are, because choosing among them requires the learner to consider perceptual accuracy and what the reader is being asked to compare. Assembling a set of this kind requires knowing which variations in a specification are consequential for the dataset at hand, which cannot be determined in advance of seeing the data. Surfacing alternatives has been proposed before as a way to counter design fixation [19], but not what makes an alternative worth considering.

Diagnostic questioning. The agent poses a question intended to locate the boundary of the learner’s reasoning. A learner may give a correct reason for a design choice without knowing how far that reason extends. A counterfactual reveals the extent: does the same reason still hold if the reader’s question changes? The agent needs to know what else this dataset could be asked to show. That is a judgment about this particular dataset, not a rule that applies to bar charts in general. Critique-first workflows ask a learner to evaluate an output before the system reveals its own assessment [11]. Diagnostic questioning targets a later point, after a learner has committed to a choice and given a reason for it. Section 5 shows this in the relating-to-a-task step.

Rationale checking. The agent reads what the learner has written and identifies what it omits. A rationale that cites data type but says nothing about the intended audience, for example, is incomplete in a specific, nameable way. Naming that omission is more useful to the learner than a score, since it requires holding the stated reason against the submitted design and identifying the decisions the reason does not cover. Justification prompts require a learner to state a reason before adopting an output [23]. This judgment can apply at either position, as a gate that withholds adoption until a reason is given, or as a check on a reason already given, or as a check on a reason already given.

Table 2: Five agent skills, the friction position each delivers, and the judgment each requires.

Skill	Position	Judgment the agent must make
Intent elicitation	Pre-generation	Whether the stated intent is a claim or only a topic
Withholding	Pre-generation	Whether the learner is blocked or deferring
Alternative generation	Pre-acceptance	Which specification variations matter for this data
Diagnostic questioning	Post-acceptance	What other analytic task this data supports
Rationale checking	Pre & Post-acceptance	Which design decisions the stated reason leaves uncovered

5 A TEACHING CASE: REPAIRING A MISLEADING CHART

Our example is an exercise suited to introductory visualization teaching. Students are given the bar chart in Figure 2a, showing quarterly sales figures of 92, 94, 95, and 97 on a y-axis beginning at 90. The truncation makes the final quarter read as more than three times the height of the first, though the underlying difference is five percent. Students are asked to critically analyze the design and redesign the visualization so that it represents the data honestly.

The exercise has no single correct answer, which is why we use it here. Truncating the axis violates the convention that bar length is proportional to value, but which repair is appropriate depends on the reader’s question. A zero baseline restores proportionality and in doing so compresses the differences to near-invisibility (Figure 2b). A line chart moves the quantity onto position, where small changes remain legible, at the cost of implying continuity between discrete periods. An explicit axis break retains the original framing and marks it. Any of the three is defensible given a reason, and it is the reason rather than the chart that we are trying to develop.

5.1 Without friction

A student using a current GenAI tool can complete this exercise in one exchange: the chart is uploaded with a request to fix it, the tool identifies the truncated axis, and a zero-baseline version is returned. The submitted work is indistinguishable from that of a student who reasoned it through. Four decisions were made by the tool. The student did not critically analyze why the original misleads, did not learn that more than one repair exists, did not relate the choice of repair to a reader’s task, and did not commit to a rationale that could be examined.

5.2 With friction

We now describe the same exercise mediated by an agent that introduces friction at each of these four points.

Diagnosis (intent elicitation, pre-generation). Before offering any analysis, the agent asks what a reader would conclude from the chart as it stands. The student answers that the final quarter appears much larger. The agent asks for the actual values, and the student reads them off and sees the discrepancy. The proportionality problem is one the student states rather than receives.

Generating repairs (alternative generation, pre-acceptance). The agent returns three charts instead of one: a zero baseline, a line chart, and an explicit axis break. The three differ in which visual channel carries the quantity, so choosing among them requires reasoning about perceptual accuracy rather than about appearance. The student is asked to choose and to give the basis for the choice.

Relating the choice to a task (diagnostic questioning, post-acceptance). The student selects the zero baseline on the grounds that bar length should be proportional to value. The agent accepts this and then asks whether the same chart would serve a reader whose question is whether growth is accelerating. The point of the

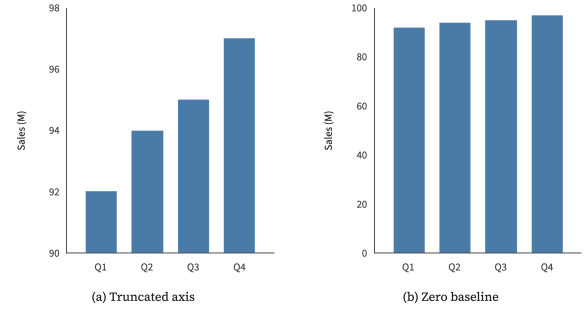


Figure 2: The same quarterly sales figures under a truncated axis (a) and a zero baseline (b). The truncated axis makes the Q4 bar read as more than three times the height of Q1, though the actual difference is five percent.

question is that the choice is task-relative, and that a rule correctly applied in one case may not transfer.

Committing to a rationale (rationale checking, post-acceptance). The student writes a justification. The agent reads it and names what it does not address: the appeal to proportionality is present, but the intended audience is not. The student is asked to identify the audience and revise accordingly.

5.3 What the exercise shows

Each of these four points is a juncture at which a decision admits more than one defensible answer, and it is at such junctures that delegation to a capable agent removes the reasoning the exercise was designed to develop. This suggests where friction belongs in visualization design learning, which is at the points where the design space branches rather than at the points where a rule is applied. The exercise illustrates all three friction positions from Section 3 and four of the five skills from Section 4. The remaining skill, withholding, and the further frictions available at each position are drawn from the literature in Section 2 rather than demonstrated here.

6 DISCUSSION AND FUTURE WORK

Section 3.2 established that an upper bound exists without giving any means of locating it, and the most convenient method is unavailable, since learners prefer the interfaces on which they perform worst [5]. A workable answer would vary friction with the learner’s current capability. That converts calibration from a parameter choice into an adaptivity problem: can an agent assess that capability from their design decisions, without a separate test?

We have not shown that an agent with these skills actually preserves design reasoning. Showing that would mean comparing learners who work with such an agent to learners who do not, on tasks where we can see the reasoning itself, not just infer it from the submitted chart. Doing that first requires deciding what counts as evidence of design reasoning, in a task that has no correct answer to check it against. What should be evaluated is not the final chart but the trace of exchanges that produced it.

Two directions follow. An observational study of how experienced instructors introduce friction in visualization design teaching would test whether these five skills describe what human tutors do, and would very likely extend the set. An implementation is also within reach. The skills can be delivered as a layer over an existing conversational interface, requiring no new authoring tool.

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